Assignment 3

Files Manipulation

AUTHOR
Operating Systems Teaching and Research Unit

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Assignment

Tee system call

In programming, especially when working with files, you may encounter the need to duplicate data to multiple destinations. A common scenario is application logging, where you want to view logs in real-time on the terminal and also save them to a file for later analysis. The `tee` function provides a convenient way to achieve this.

Implement the `tee` function. It reads the standard input and write it back both to:

1. the standard output
2. the file taken in parameter

```c
int tee(char *output_file);
```

The standard input is represented by the file descriptor `0`. You can also use the macro `STDIN_FILENO`. The same applies for the standard output, where the macro `STDOUT_FILENO` is equal to `1`.

The `output_file` **must** be created with permissions `rw-r--r--`.

The function shall return 0 in case of a success, and -1 in case of an error.

Read the manual

This function is inspired by the command `tee`. Check the manual: `man tee(1)`.

Reverse the bytes of a file

Reverse the content of `input_file` byte by byte. You are expected to use `lseek` to achieve this. Write your output to `output_file`.

```c
int reverse_file(char *input_file, char *output_file);
```

The `output_file` **must** be created with read/write permissions.

The function shall return 0 in case of a success, and -1 in case of an error.

Optimize your code

Optimize `reverse_file`.

This last function is deliberately harder than the rest of this assignment. Calling read/write on every byte can be very inefficient. One way to optimize the code is to use a temporary buffer to work on a small part of the file at a time.

```
int reverse_file_optimized(char *input_file, char *output_file);
```

You will need to be careful not using to much resources in terms of:

1. memory
2. execution time

If the program is taking too much time or too much memory, it will crash.

> **Remark**
>
> If you end up with this kind of output, it means that the program has been killed because it took too long to execute.
>
> ```
> Test 5: Bigfile reverse test
> Configuration error in the test case: the output is not definedProgram timeout
> Incorrect exit code. Expected 0, found -2147483648
> Incorrect program output
> --- Input ---
>
> --- Program output ---
> ```